DATA 260 – Homework 8 Report

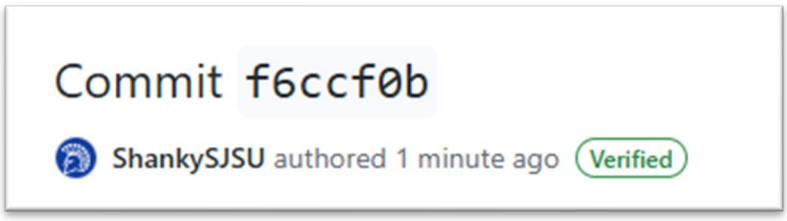
Student ID: 018322837 **Name: Shashank Ranjan**

Section 0 – Personal Configuration and Domain

Name	Definition	Value
SID	Student ID	018322837
SID4	Last 4 of Student ID	2837
PORT_BASE	8000 + SID mod 900(137)	8137
prefix	"s" + SID4	s2837
Seed	SID4	2837
verify sid	260000 + SID4	262837
DOMAIN_ID	SID4 MOD 8	5
Assigned Domain	Assignment domain	**Local restaurant inspections**

Repository: https://github.com/ShankySJSU/Data260-2837

Submission Tag: hw8 (how to create tag 😊 **Tagged commit hash (hw8):**



f6ccf0be3bca53431c4578df5a0a09e4bc383f35

Pleae NOTE this could be different in the WORD document that you find on the repository, as this is source of the PDF file that I will be uploading to Canvas and the Report

(Shashank: always use "git rev-parse HEAD"

Collaborators with access: Sbnikitha, supriyaselvanganesan

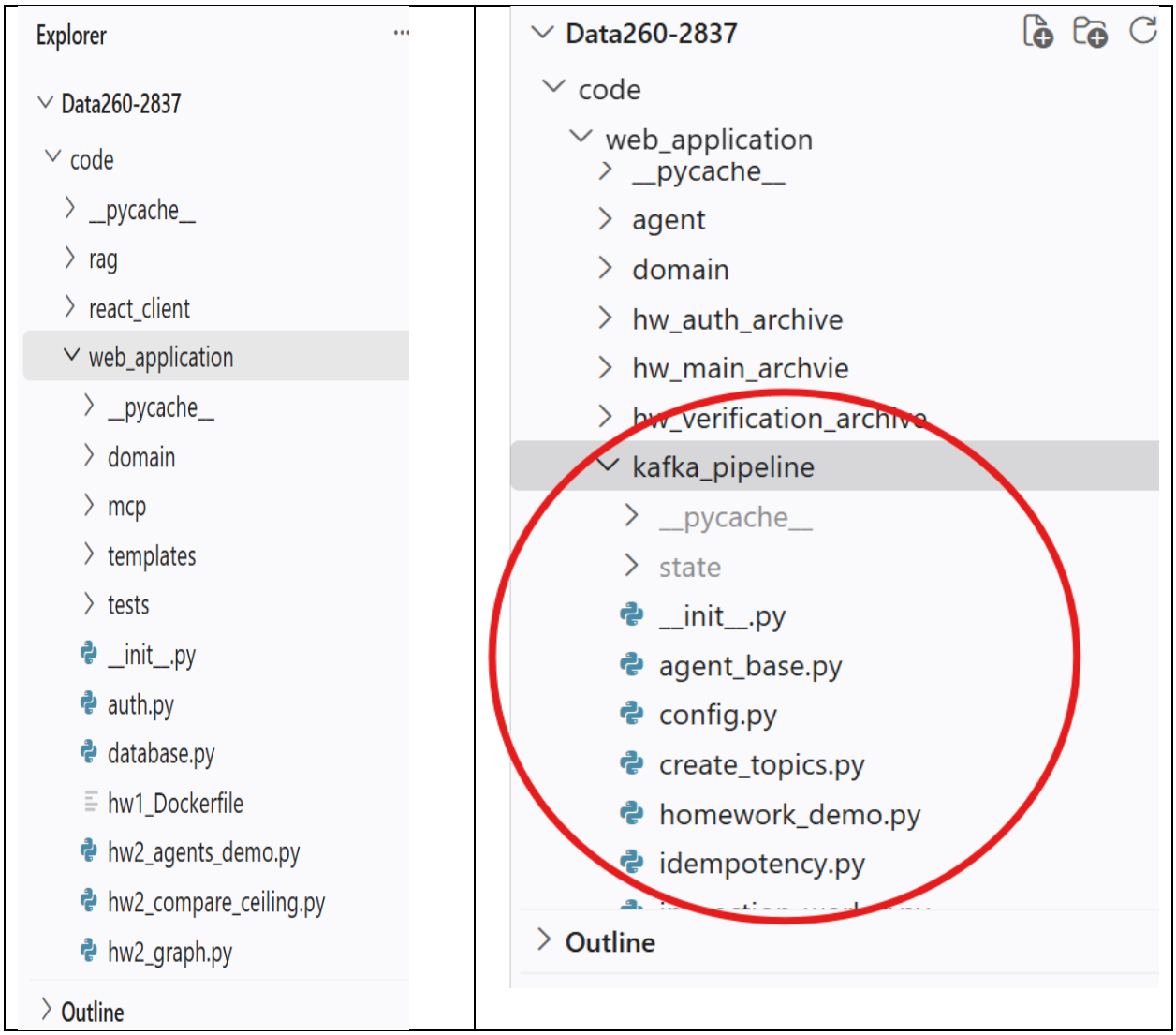
Hardware:

Local Windows PC, Python 3.12, VS Code Desktop, i7 CPU, 16GB RAM.

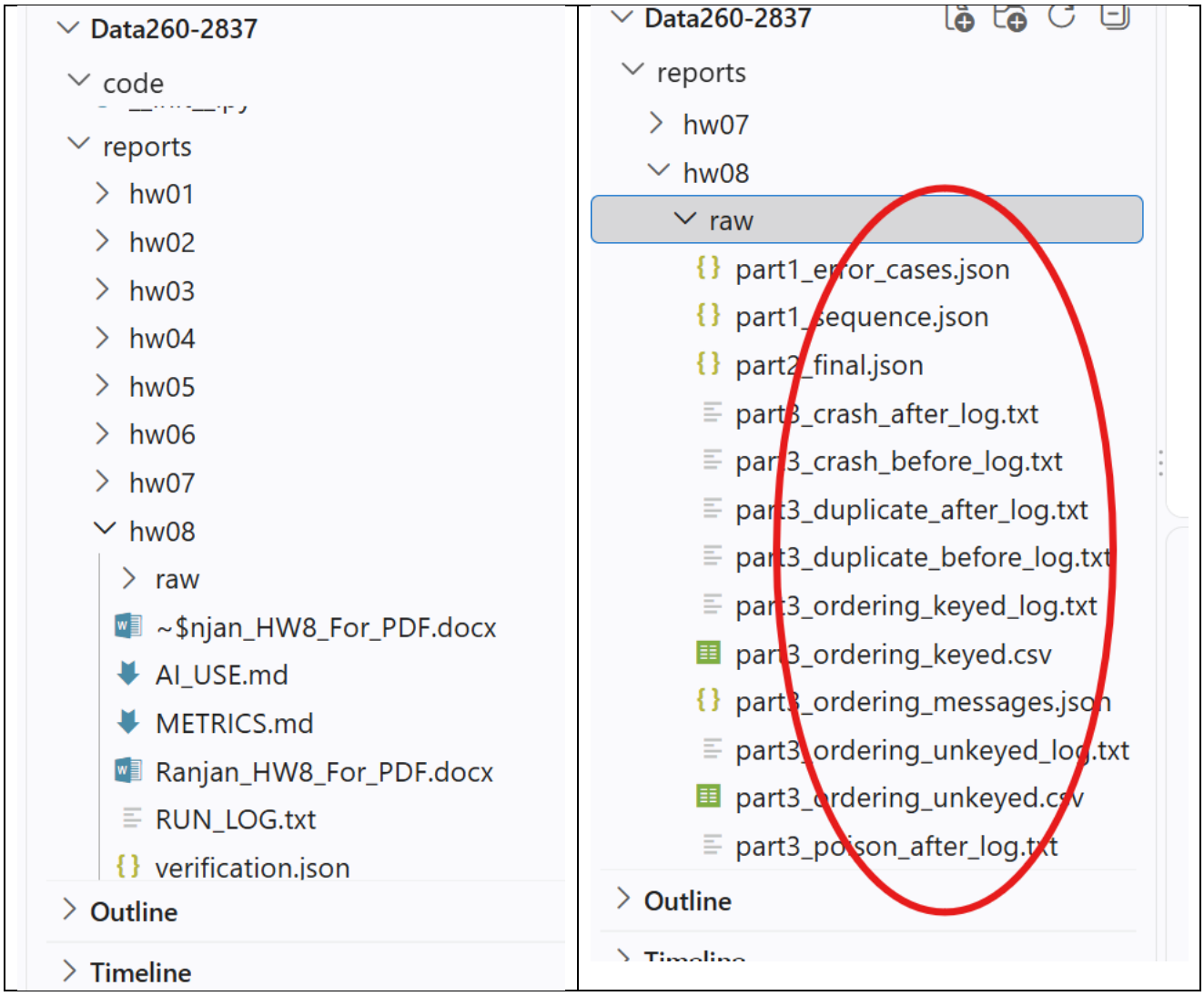
Model Used:

- Embeddings: sentence-transformers/all-MiniLM-L6-v2
- Ollama (qwen2.5:1.5b, qwen3:4b, llama3.2:1b)

Base Directory Structure:



New and changed HW8 files are boxed in red. The code is an extension of the existing HW1–HW7 codebase (no separate copy).



Part 1 - Backend Service Request/Response over Kafka

The domain logic for my primary entity (Inspection) now runs in a separate worker process. The client / FastAPI wrapper and the worker communicate only through two Kafka topics; the worker keeps inspections in a plain Python dictionary and validates input with my existing InspectionCreate Pydantic schema.

1.1 Topics and operations

Topic	Written by	Read by	Partitions
s2837_inspection_requests	client / FastAPI (hw8_main.py)	inspection_worker.py	1
s2837_inspection_responses	inspection_worker.py (reply_to)	client / FastAPI	1

Operation	Input data	Success output	Error output
CREATE_INSPECTION	name, inspection_code,	{"success": true,	{"success": false, "error":

	inspection_date, status, score, restaurant_id	"inspection_id": "<uuid4>", "message": "Inspection created"}	"Invalid inspection data - ..."}
GET_INSPECTION	{"inspection_id": "<uuid>"}	{"success": true, "inspection": {...}}	{"success": false, "error": "Inspection not found"}
anything else	-	-	{"success": false, "error": "Unknown operation"}

1.2 Request and response message schemas

```
REQUEST  (s2837_inspection_requests)
{
  "correlation_id": "81382958-c78b-454f-af44-b8ea73093926",
  "reply_to": "s2837_inspection_responses",
  "operation": "CREATE_INSPECTION",
  "data": {
    "name": "Rangoli Indian Cuisine - routine inspection",
    "inspection_code": "S2837-INSP-0001",
    "inspection_date": "2026-11-02T10:30:00",
    "status": "PASS",
    "score": 94,
    "restaurant_id": 1
  }
}

RESPONSE (s2837_inspection_responses)
{
  "correlation_id": "81382958-c78b-454f-af44-b8ea73093926",
  "operation": "CREATE_INSPECTION",
  "success": true,
  "inspection_id": "5c72cee2-0d9a-4186-adaa-8705114cde4b",
  "message": "Inspection created",
  "worker_processed_at": "2026-11-02T07:46:13.879+00:00"
}
```

1.3 correlation_id flow

- The client (KafkaRequestClient.request / homework_demo.py) creates the correlation_id once with uuid4 and puts it in the request together with reply_to, operation and data.
- FastAPI accepts an X-Correlation-ID header and forwards it unchanged (it only creates one when the caller did not).
- The worker never generates a new id: it copies correlation_id into every response, including error responses, and publishes to the reply_to topic.
- The client's background listener matches the response to the waiting request by correlation_id. A repeated correlation_id returns the stored reply (no second entity).

1.4 Code - worker, client and API wrapper

```
# inspection_worker.py
def process(request):
    """Validate the envelope and run the operation. Always returns a response dict."""
    correlation_id = request.get("correlation_id")
    operation = request.get("operation")
```

```
    if not isinstance(correlation_id, str) or not correlation_id:
        body = {"success": False, "error": "Missing correlation_id"}
    elif "data" not in request:
        body = {"success": False, "error": "Missing data"}
    elif operation not in HANDLERS:
        body = {"success": False, "error": "Unknown operation"}
    else:
        body = HANDLERS[operation](request.get("data"))
    return {"correlation_id": correlation_id, "operation": operation, **body,
            "worker_processed_at": now_iso()}
```

```
# request_client.py
def request(self, operation, data, correlation_id=None, timeout=15):
        message = self.build_request(operation, data, correlation_id)
        correlation_id = message["correlation_id"]
        slot = {"event": threading.Event()}
        with self._lock:
            self._pending[correlation_id] = slot
        started = time.perf_counter()
        meta = send_json(self.producer, cfg.REQUEST_TOPIC, message, key=correlation_id)
        log(self.component, f"SENT to {cfg.REQUEST_TOPIC}[p{meta.partition}@{meta.offset}] "
                            f"correlation_id={correlation_id} message={short(message)}")
        trace(self.component, "CLIENT_SEND", correlation_id, operation=operation,
              topic=cfg.REQUEST_TOPIC, partition=meta.partition, offset=meta.offset,
              message=message)
        try:
            if not slot["event"].wait(timeout):
                raise TimeoutError(f"no response for correlation_id={correlation_id} "
                                   f"within {timeout}s - is the worker running?")
        finally:
            with self._lock:
                self._pending.pop(correlation_id, None)
        response = slot["response"]
        elapsed_ms = round((time.perf_counter() - started) * 1000, 1)
    # ... (see repository for the rest)
```

```
# hw8_main.py
async def create_inspection_kafka(data: dict = Body(...),
                                  x_correlation_id: str | None = Header(default=None)):
    return await kafka_call(cfg.OP_CREATE, data, x_correlation_id or str(uuid.uuid4()))


# hw8_main.py
async def kafka_call(operation, data, correlation_id):
    log(COMPONENT, f"HTTP -> Kafka operation={operation} correlation_id={correlation_id}")
    try:
        message, response = await run_in_threadpool(
            STATE["client"].request, operation, data, correlation_id)
    except TimeoutError as exc:
        raise HTTPException(status_code=504, detail=str(exc)) from exc
    log(COMPONENT, f"Kafka -> HTTP returning to caller correlation_id="
                   f"{response.get('correlation_id')} success={response.get('success')}")
    return {"request": message, "response": response}
```

1.5 screenshots

Step 1 : **docker compose up -d --build**

```
C:\SJSURepo\Data260\Data260-2837>echo "Shashank -HW 8 - Starting Docker"
"Shashank -HW 8 - Starting Docker"

C:\SJSURepo\Data260\Data260-2837>docker compose up -d --build
[+] Building 27.7s (21/21) FINISHED
```

```
 => [internal] load local bake definitions                                                                    0.0s
 => => reading from stdin 975B                                                                                0.0s
 => [api internal] load build definition from Dockerfile.hw8                                                  0.0s
 => => transferring dockerfile: 954B                                                                          0.0s
 => [worker internal] load metadata for docker.io/library/python:3.12-slim                                    1.4s
 => [api internal] load .dockerignore                                                                         0.0s
 => => transferring context: 153B                                                                             0.0s
 => [worker internal] load build context                                                                      0.0s
 => => transferring context: 1.42kB                                                                           0.0s
 => [worker  1/12] FROM docker.io/library/python:3.12-slim@sha256:aa943ce411423c7582288f18707ff66eddcbaeb510502d7  3.7s
 => => resolve docker.io/library/python:3.12-slim@sha256:aa943ce411423c7582288f18707ff66eddcbaeb510502d747a72aeb3  0.0s
 => => sha256:acca7542dfc736e9ce20cbefc58b2d09822276d15d4a0c531531a6f1b057928e 251B / 251B                    0.1s
 => => sha256:f574edac59d9f092df596f54e5b52656c803fd3ae30825b8ee5a189ed05f63d3 12.12MB / 12.12MB              1.0s
 => => sha256:3eb771151506765 68f88e2957181b8f7e3014e797f1ce8cd640086dfbed45e35 1.29MB / 1.29MB               0.5s
 => => sha256:6fc242d221a673263eabd14a9ad9584d3177c96e5cb504c891497338e2f912f5 29.86MB / 29.86MB              1.7s
 => => extracting sha256:6fc242d221a673263eabd14a9ad9584d3177c96e5cb504c891497338e2f912f5                      1.0s
 => => extracting sha256:3eb771151506765 68f88e2957181b8f7e3014e797f1ce8cd640086dfbed45e35                    0.1s
 => => extracting sha256:f574edac59d9f092df596f54e5b52656c803fd3ae30825b8ee5a189ed05f63d3                      0.6s
 => => extracting sha256:acca7542dfc736e9ce20cbefc58b2d09822276d15d4a0c531531a6f1b057928e                      0.0s
 => [worker  2/12] WORKDIR /app/code                                                                          0.2s
 => [worker  3/12] RUN pip install kafka-python>=2.1.0                                                        2.7s
 => [worker  4/12] RUN pip install fastapi>=0.110                                                             5.4s
 => [api  5/12] RUN pip install uvicorn>=0.29                                                                 2.0s
 => [worker  6/12] RUN pip install pydantic>=2.5                                                              0.8s
 => [worker  7/12] RUN pip install requests>=2.31                                                             2.1s
 => [api  8/12] RUN pip install python-docx>=1.1                                                              2.9s
 => [api  9/12] COPY code/web_application/__init__.py /app/code/web_application/__init__.py                   0.1s
```

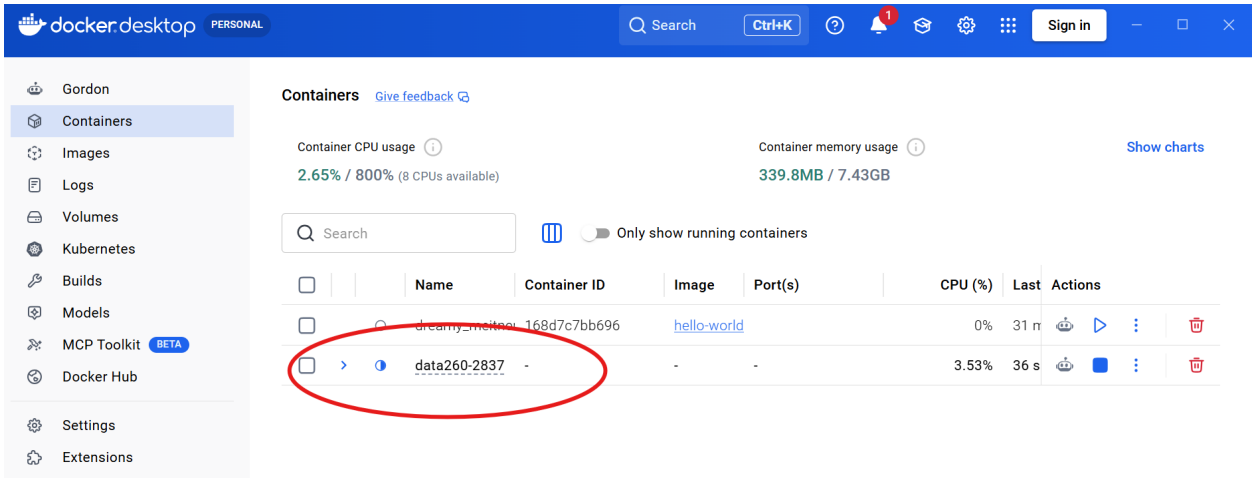
```
=> [worker  7/12] RUN pip install requests>=2.31                                                             2.1s
=> [api  8/12] RUN pip install python-docx>=1.1                                                              2.9s
=> [api  9/12] COPY code/web_application/__init__.py /app/code/web_application/__init__.py                   0.1s
=> [worker 10/12] COPY code/web_application/hw8_main.py /app/code/web_application/hw8_main.py                0.1s
=> [worker 11/12] COPY code/web_application/domain/__init__.py code/web_application/domain/schema.py /app/code/w  0.1s
=> [api 12/12] COPY code/web_application/kafka_pipeline /app/code/web_application/kafka_pipeline             0.1s
=> [api] exporting to image                                                                                  5.0s
=> => exporting layers                                                                                       1.5s
=> => exporting manifest sha256:194288ad4c6b17f8f1ace18852276f0d4765bc5788893c755166c8da414ff289             0.0s
=> => exporting config sha256:06dc5fad84fe33c3dc056c32e57e448f5ba1716fe379af86883cf3e393832c84               0.0s
=> => exporting attestation manifest sha256:47f435e9ff28702db308cb53a57687604f4ba885c300d2cf2d2d9e2692596fb0  0.1s
=> => exporting manifest list sha256:d3680e49f47c43549fdfd2f1d7f69ae2862cac87d3547ff59ef2c75ca4687aed         0.0s
=> => naming to docker.io/library/data260-2837-api:latest                                                    0.0s
=> => unpacking to docker.io/library/data260-2837-api:latest                                                 3.3s
=> [worker] exporting to image                                                                               5.0s
=> => exporting layers                                                                                       1.5s
=> => exporting manifest sha256:04a5e4558650dbe4d1da87518460be61712149d8f9bca2412fc0f1330b9044d2             0.0s
=> => exporting config sha256:ecc0a5f1b24d645fa894dc867fd4a804179ac4bbbd047b600b8bd5dbbb2d8be0               0.0s
=> => exporting attestation manifest sha256:bc22bf8536d5f8754856b51ef754fe3f19e61d10780c55f7ed88dd87856c7dca  0.1s
=> => exporting manifest list sha256:930869f1a28446238aac602b467c7c8c4239c03342ac7e2a4a18aa180ce543a5        0.0s
=> => naming to docker.io/library/data260-2837-worker:latest                                                 0.0s
=> => unpacking to docker.io/library/data260-2837-worker:latest                                              3.3s
=> [api] resolving provenance for metadata file                                                              0.0s
=> [worker] resolving provenance for metadata file                                                           0.0s
[+] up 6/6
 ✔Image data260-2837-worker    Built                                                                        27.9s
 ✔Image data260-2837-api       Built                                                                        28.0s
 ✔Network data260-2837_default Created                                                                       0.0s
 ✔Container s2837-kafka        Healthy                                                                      13.9s
 ✔Container s2837-worker       Started                                                                      13.1s
 ✔Container s2837-api          Started                                                                      12.9s

C:\SJSURepo\Data260\Data260-2837>
```

C:\SJSURepo\Data260\Data260-2837>docker ps

```
C:\SJSURepo\Data260\Data260-2837>docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                            PORTS
                        NAMES
0d885033824c   data260-2837-api       "python -m uvicorn w…"   4 minutes ago   Restarting (1) 2 seconds ago
                        s2837-api
3012926c55b3   data260-2837-worker    "python web_applicat…"   4 minutes ago   Restarting (1) 5 seconds ago
                        s2837-worker
c04b4bd68d42   apache/kafka:3.8.0     "/__cacert_entrypoin…"   4 minutes ago   Up 4 minutes (healthy)            0.0.0.0:909
2->9092/tcp, [::]:9092->9092/tcp   s2837-kafka

C:\SJSURepo\Data260\Data260-2837>
```



C:\SJSURepo\Data260\Data260-2837>**docker logs s2837-worker**

```
C:\SJSURepo\Data260\Data260-2837>docker logs s2837-worker
2026-11-02 07:14:51.474 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_inspection_requests (partitions=1)
2026-11-02 07:14:51.513 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_inspection_responses (partitions=1)
2026-11-02 07:14:51.550 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_inbox (partitions=1)
2026-11-02 07:14:51.589 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_tasks (partitions=2)
2026-11-02 07:14:51.628 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_drafts (partitions=1)
2026-11-02 07:14:51.667 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_final (partitions=1)
2026-11-02 07:14:51.706 | Shashank Ranjan | s2837 | WORKER     | created topic s2837_dead_letter (partitions=1)
2026-11-02 07:14:51.722 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_inspection_requests partitions=1
2026-11-02 07:14:51.728 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_inspection_responses partitions=1
2026-11-02 07:14:51.729 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_inbox partitions=1
2026-11-02 07:14:51.732 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_tasks partitions=2
2026-11-02 07:14:51.734 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_drafts partitions=1
2026-11-02 07:14:51.736 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_final partitions=1
2026-11-02 07:14:51.739 | Shashank Ranjan | s2837 | WORKER     | topic ready s2837_dead_letter partitions=1
2026-11-02 07:14:51.754 | Shashank Ranjan | s2837 | WORKER     | connecting to kafka:29092, group=s2837_inspection_worke
rs
2026-11-02 07:14:53.761 | Shashank Ranjan | s2837 | WORKER     | Consumer is ready and listening... topic=s2837_inspecti
on_requests partitions=[0]

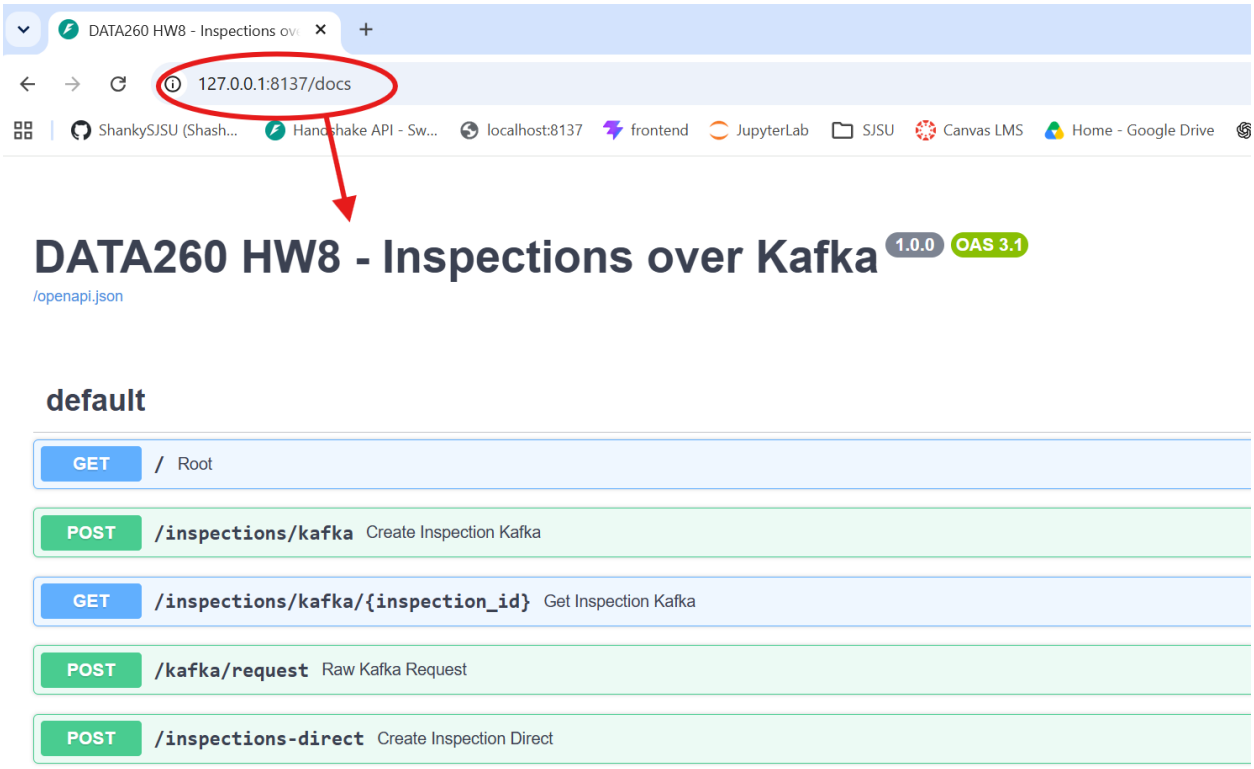
C:\SJSURepo\Data260\Data260-2837>
```

C:\SJSURepo\Data260\Data260-2837>docker logs s2837-api

```
C:\SJSURepo\Data260\Data260-2837>docker logs s2837-api
INFO:     Started server process [1]
INFO:     Waiting for application startup.
2026-11-02 07:14:51.727 | Shashank Ranjan | s2837 | API        | topic ready s2837_inspection_requests partitions=1
2026-11-02 07:14:51.730 | Shashank Ranjan | s2837 | API        | topic ready s2837_inspection_responses partitions=1
2026-11-02 07:14:51.732 | Shashank Ranjan | s2837 | API        | topic ready s2837_inbox partitions=1
2026-11-02 07:14:51.734 | Shashank Ranjan | s2837 | API        | topic ready s2837_tasks partitions=2
2026-11-02 07:14:51.736 | Shashank Ranjan | s2837 | API        | topic ready s2837_drafts partitions=1
2026-11-02 07:14:51.738 | Shashank Ranjan | s2837 | API        | topic ready s2837_final partitions=1
2026-11-02 07:14:51.740 | Shashank Ranjan | s2837 | API        | topic ready s2837_dead_letter partitions=1
2026-11-02 07:14:51.766 | Shashank Ranjan | s2837 | API        | FastAPI server started at http://127.0.0.1:8137 (Kafka
kafka:29092, requests=s2837_inspection_requests, responses=s2837_inspection_responses)
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8137 (Press CTRL+C to quit)

C:\SJSURepo\Data260\Data260-2837>
```

Browser: http://127.0.0.1:8137/docs



Topics (worker/API already create them; this prints the CLI form and confirms):

C:\SJSURepo\Data260\Data260-2837>**cd code**

C:\SJSURepo\Data260\Data260-2837\code>python web_application\kafka_pipeline\create_topics.py

```
C:\SJSURepo\Data260\Data260-2837\code>python web_application\kafka_pipeline\create_topics.py
2026-11-01 23:24:48.763 | Shashank Ranjan | s2837 | SETUP      | equivalent CLI (inside the kafka container):
2026-11-01 23:24:48.764 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_inspection_requests --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.766 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_inspection_responses --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.767 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_inbox --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.767 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_tasks --partitions 2 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.767 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_drafts --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.769 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_final --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.770 | Shashank Ranjan | s2837 | SETUP      |   /opt/kafka/bin/kafka-topics.sh --create --if-not-exis
ts --topic s2837_dead_letter --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
2026-11-01 23:24:48.798 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_inspection_requests partitions=1
2026-11-01 23:24:48.799 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_inspection_responses partitions=1
2026-11-01 23:24:48.799 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_inbox partitions=1
2026-11-01 23:24:48.800 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_tasks partitions=2
2026-11-01 23:24:48.800 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_drafts partitions=1
2026-11-01 23:24:48.802 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_final partitions=1
2026-11-01 23:24:48.802 | Shashank Ranjan | s2837 | SETUP      | topic ready s2837_dead_letter partitions=1

C:\SJSURepo\Data260\Data260-2837\code>
```

C:\SJSURepo\Data260\Data260-2837\code>echo "Shashank Ranjan -hW 8 - Demo using API"

"Shashank Ranjan -hW 8 - Demo using API"

C:\SJSURepo\Data260\Data260-2837\code>**python web_application\kafka_pipeline\homework_demo.py --via-api**

2026-11-01 23:41:16.099 | Shashank Ranjan | s2837 | DEMO |
==========

2026-11-01 23:41:16.101 | Shashank Ranjan | s2837 | DEMO | HW8 Part 1 demo - Shashank Ranjan - SID4=2837 PREFIX=s2837 - via FastAPI

```
t-response  API     s2837_inspection_responses[p0@0] correlation_id=73a0a127-79a0-4ef4-9c59-7d22c8f1238d
2026-11-01 23:41:17.842 | Shashank Ranjan | s2837 | DEMO       | 2026-11-02T07:41:16.274+00:00  GET_INSPECTION     send
           API     s2837_inspection_requests[p0@1] correlation_id=0df0afbf-1018-44a3-8c9e-498e16d4111a
2026-11-01 23:41:17.843 | Shashank Ranjan | s2837 | DEMO       | 2026-11-02T07:41:16.278+00:00  GET_INSPECTION     recei
ve          WORKER  s2837_inspection_requests[p0@1] correlation_id=0df0afbf-1018-44a3-8c9e-498e16d4111a
2026-11-01 23:41:17.844 | Shashank Ranjan | s2837 | DEMO       | 2026-11-02T07:41:16.287+00:00  GET_INSPECTION     proce
ss          WORKER   correlation_id=0df0afbf-1018-44a3-8c9e-498e16d4111a
2026-11-01 23:41:17.845 | Shashank Ranjan | s2837 | DEMO       | 2026-11-02T07:41:16.310+00:00  GET_INSPECTION     clien
t-response  API     s2837_inspection_responses[p0@1] correlation_id=0df0afbf-1018-44a3-8c9e-498e16d4111a
2026-11-01 23:41:17.846 | Shashank Ranjan | s2837 | DEMO       | note: worker trace lines missing - is reports/ mounted
into the worker?
2026-11-01 23:41:17.850 | Shashank Ranjan | s2837 | DEMO       | saved C:\SJSURepo\Data260\Data260-2837\reports\hw08\raw
\part1_sequence.json
2026-11-01 23:41:17.852 | Shashank Ranjan | s2837 | DEMO       | ========================================================
==============
2026-11-01 23:41:17.852 | Shashank Ranjan | s2837 | DEMO       | ERROR CASES (each must return success=false + same corr
elation_id)
2026-11-01 23:41:17.854 | Shashank Ranjan | s2837 | DEMO       | ========================================================
==============
2026-11-01 23:41:17.931 | Shashank Ranjan | s2837 | DEMO       | unknown inspection_id    -> success=False error='Inspec
tion not found' correlation_id_preserved=True
2026-11-01 23:41:17.981 | Shashank Ranjan | s2837 | DEMO       | unknown operation        -> success=False error='Unknow
n operation' correlation_id_preserved=True
2026-11-01 23:41:18.034 | Shashank Ranjan | s2837 | DEMO       | missing/invalid fields   -> success=False error='Invali
d inspection data - name: String should have at least 2 characters; inspection_code: String should have at least 8 chara
cters; inspection_date: Field required; status: Value error, status must be PASS, FAIL, or WARNING; score: Input should
be less than or equal to 100; restaurant_id: Field required' correlation_id_preserved=True
2026-11-01 23:41:18.037 | Shashank Ranjan | s2837 | DEMO       | demo finished - all checks passed: True

C:\SJSURepo\Data260\Data260-2837\code>
```

code> python web_application\kafka_pipeline\homework_demo.

```
C:\SJSURepo\Data260\Data260-2837\code>echo "Shashank Ranjan -hW 8 - Demo"
"Shashank Ranjan -hW 8 - Demo"

C:\SJSURepo\Data260\Data260-2837\code>python web_application\kafka_pipeline\homework_demo.py
2026-11-01 23:46:13.697 | Shashank Ranjan | s2837 | DEMO       | ========================================================
==============
2026-11-01 23:46:13.697 | Shashank Ranjan | s2837 | DEMO       | HW8 Part 1 demo - Shashank Ranjan - SID4=2837 PREFIX=s2
837 - Kafka client
2026-11-01 23:46:13.699 | Shashank Ranjan | s2837 | DEMO       | ========================================================
==============
2026-11-01 23:46:13.736 | Shashank Ranjan | s2837 | DEMO       | STEP 1  created correlation_id=81382958-c78b-454f-af44-
b8ea73093926
2026-11-01 23:46:13.737 | Shashank Ranjan | s2837 | DEMO       | STEP 2  sending CREATE_INSPECTION to s2837_inspection_r
equests
2026-11-01 23:46:13.858 | Shashank Ranjan | s2837 | CLIENT     | SENT to s2837_inspection_requests[p0@5] correlation_id=
81382958-c78b-454f-af44-b8ea73093926 message={"correlation_id": "81382958-c78b-454f-af44-b8ea73093926", "reply_to": "s28
37_inspection_responses", "operation": "CREATE_INSPECTION", "data": {"name": "Rangoli Indian Cuisine - routine inspectio
n", "inspection_code": "S2837-INSP-0001", "inspection_date": "2026-11-02T10:30:00", "status": "PASS", "score": 94, "rest
aurant_id": 1}}
2026-11-01 23:46:13.893 | Shashank Ranjan | s2837 | CLIENT     | RECEIVED response from s2837_inspection_responses[p0@5]
```

1.6 Full sequence CREATE_INSPECTION -> GET_INSPECTION (from raw/part1_sequence.json)

```
07:46:13.859  CREATE_INSPECTION  CLIENT_SEND     CLIENT  s2837_inspection_requests[p0@5] cid=81382958
07:46:13.868  CREATE_INSPECTION  WORKER_RECEIVE  WORKER  s2837_inspection_requests[p0@5] cid=81382958
07:46:13.879  CREATE_INSPECTION  WORKER_PROCESS  WORKER   cid=81382958
07:46:13.899  CREATE_INSPECTION  WORKER_REPLY    WORKER  s2837_inspection_responses[p0@5] cid=81382958
07:46:13.895  CREATE_INSPECTION  CLIENT_RECEIVE  CLIENT  s2837_inspection_responses[p0@5] cid=81382958
07:46:13.905  GET_INSPECTION     CLIENT_SEND     CLIENT  s2837_inspection_requests[p0@6] cid=6e6c3788
07:46:13.912  GET_INSPECTION     WORKER_RECEIVE  WORKER  s2837_inspection_requests[p0@6] cid=6e6c3788
07:46:13.922  GET_INSPECTION     WORKER_PROCESS  WORKER   cid=6e6c3788
07:46:13.940  GET_INSPECTION     WORKER_REPLY    WORKER  s2837_inspection_responses[p0@6] cid=6e6c3788
07:46:13.936  GET_INSPECTION     CLIENT_RECEIVE  CLIENT  s2837_inspection_responses[p0@6] cid=6e6c3788
```

Matching console lines from RUN_LOG.txt (client + worker):

```
23:46:13.858 CLIENT      SENT to s2837_inspection_requests[p0@5] correlation_id=81382958-c78b-454f-af44-
b8ea73093926 message={"correlation_id": "81382958-c78b-454f-af4...
07:46:13.858 WORKER      RECEIVED from s2837_inspection_requests[p0@5] correlation_id=81382958-c78b-454f-
af44-b8ea73093926 operation=CREATE_INSPECTION message={"correl...
07:46:13.874 WORKER      PROCESSING CREATE_INSPECTION correlation_id=81382958-c78b-454f-af44-b8ea73093926
07:46:13.884 WORKER      PROCESSED CREATE_INSPECTION correlation_id=81382958-c78b-454f-af44-b8ea73093926
success=True inspection_id=5c72cee2-0d9a-4186-adaa-8705114cde4b
23:46:13.893 CLIENT      RECEIVED response from s2837_inspection_responses[p0@5] correlation_id=81382958-
c78b-454f-af44-b8ea73093926 success=True round_trip_ms=155.5 r...
07:46:13.894 WORKER      SENT response to s2837_inspection_responses[p0@5] correlation_id=81382958-c78b-
454f-af44-b8ea73093926 response={"correlation_id": "81382958-c7...
23:46:13.904 CLIENT      SENT to s2837_inspection_requests[p0@6] correlation_id=6e6c3788-feca-49bc-8d7d-
57abf1f3c027 message={"correlation_id": "6e6c3788-feca-49bc-8d7...
07:46:13.908 WORKER      RECEIVED from s2837_inspection_requests[p0@6] correlation_id=6e6c3788-feca-49bc-
8d7d-57abf1f3c027 operation=GET_INSPECTION message={"correlati...
07:46:13.918 WORKER      PROCESSING GET_INSPECTION correlation_id=6e6c3788-feca-49bc-8d7d-57abf1f3c027
07:46:13.926 WORKER      PROCESSED GET_INSPECTION correlation_id=6e6c3788-feca-49bc-8d7d-57abf1f3c027
success=True
23:46:13.935 CLIENT      RECEIVED response from s2837_inspection_responses[p0@6] correlation_id=6e6c3788-
feca-49bc-8d7d-57abf1f3c027 success=True round_trip_ms=34.6 re...
07:46:13.936 WORKER      SENT response to s2837_inspection_responses[p0@6] correlation_id=6e6c3788-feca-
49bc-8d7d-57abf1f3c027 response={"correlation_id": "6e6c3788-fe...
```

Result: correlation_id preserved for both requests: True

1.7 Validation and error responses

Case	success	error	correlation_id preserved
unknown inspection_id	False	Inspection not found	True
unknown operation	False	Unknown operation	True
missing/invalid fields	False	Invalid inspection data - name: String should have at least 2 characters; inspection_code: String should have at least 8 characters; inspection_date: Field required; status: Value error, status must be PASS, FAIL, or WARNING; score: Input should be less than or equal to 100; restaurant_id: Field required	True

Part 2 - Three-Agent Kafka Pipeline

Planner, Writer and Reviewer run as separate Python processes. No agent imports or calls another one - each consumes one topic and produces to the next, and every message carries the same correlation_id plus a trace list (stage, agent, pid, topic/partition/offset, timestamps).

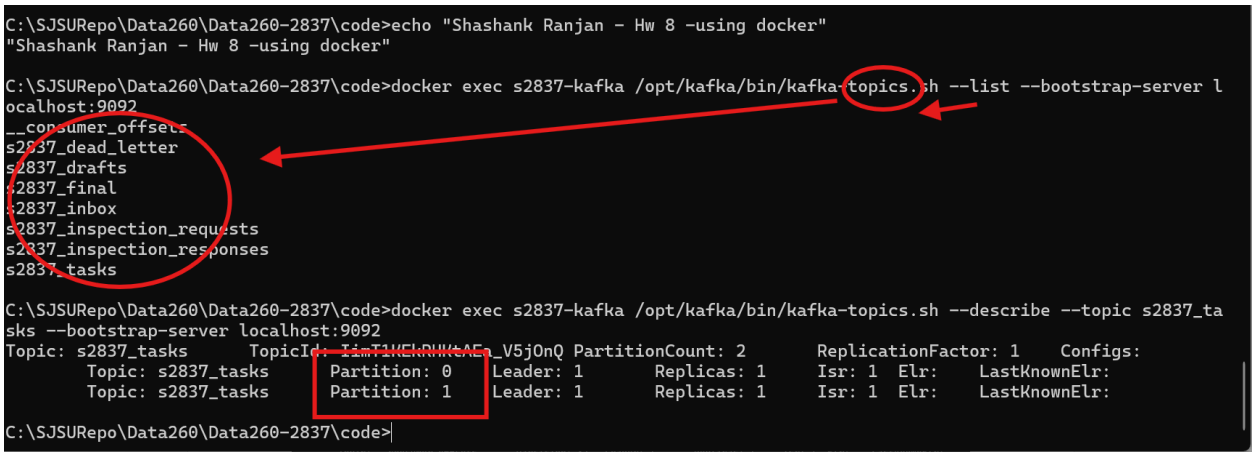
2.1 Topics and workflow

Sender	Reads topic	Sends message to
You (send_question.py)	-	s2837_inbox (a question)
Planner (planner.py)	s2837_inbox	s2837_tasks (a plan, 2 partitions)
Writer (writer.py)	s2837_tasks	s2837_drafts (a draft answer)
Reviewer (reviewer.py)	s2837_drafts	s2837_final (approval)

2.2 Topic creation

```
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic
s2837_inspection_requests --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic
s2837_inspection_responses --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s2837_inbox --
partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s2837_tasks --
partitions 2 --replication-factor 1 --bootstrap-server localhost:9092
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s2837_drafts --
partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s2837_final --
partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
docker exec s2837-kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s2837_dead_letter -
-partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
```

[SCREENSHOT - kafka-topics.sh --describe / --list showing the s2837_* topics (s2837_tasks with PartitionCount: 2)]



2.3 Agent code

```
# planner.py
def handle(message, record, runner):
    question = require(message, "question")
    steps, model = llm.plan(question, runner.component)
    log(runner.component, f"PLAN ({model}) steps={steps}")
    task = {
        "correlation_id": message["correlation_id"],
        "test_run": message.get("test_run"),
        "kind": "answer_task",
        "question": question,
        "plan": {"steps": steps, "model": model, "planned_at": now_iso()},
    }
    # key = correlation_id -> every task of one question lands in the same tasks partition
    return [(cfg.TASKS_TOPIC, task, message["correlation_id"])]
```

```
# writer.py
def handle(message, record, runner):
    kind = message.get("kind", "answer_task")
    correlation_id = message["correlation_id"]

    if kind == "ordering_event":
        entity_id = require(message, "entity_id")
        seq = message.get("seq")
        if not isinstance(seq, int):
            raise PermanentError("ordering_event needs integer seq")
        time.sleep(max(0, int(message.get("work_ms", 0))) / 1000)
        log(runner.component, f"ORDER entity={entity_id} seq={seq} partition={record.partition}"
                              f" offset={record.offset}")
        draft = {"correlation_id": correlation_id, "test_run": message.get("test_run"),
                 "kind": "ordering_result",
                 "run": message.get("run"), "entity_id": entity_id, "seq": seq,
                 "event": message.get("event"), "global_index": message.get("global_index"),
                 "partition": record.partition, "offset": record.offset,
                 "writer": runner.args.instance, "processed_at": now_iso()}
        return [(cfg.DRAFTS_TOPIC, draft, entity_id)]

    if kind != "answer_task":
        raise PermanentError(f"unknown task kind '{kind}'")
    # ... (see repository for the rest)
```

```
# reviewer.py
def review(draft):
    words = len(draft.split())
    checks = {
        "non_empty": bool(draft.strip()),
        "length_40_400_words": 40 <= words <= 400,
        "on_domain": bool(re.search(r"inspect|restaurant|food", draft, re.I)),
        "no_error_text": not re.search(r"traceback|exception|error:", draft, re.I),
    }
    return all(checks.values()), checks, words
```

2.4 Running the agents (4 terminals)

First verify .. ollam is running

```
C:\SJSURepo\Data260\Data260-2837>ollama list
NAME            ID              SIZE      MODIFIED
qwen2.5:1.5b    65ec06548149    986 MB    5 weeks ago
qwen3:4b        359d7dd4bcda    2.5 GB    5 weeks ago
llama3.2:1b     baf6a787fdff    1.3 GB    5 weeks ago

C:\SJSURepo\Data260\Data260-2837>
```

Planner is ready

```
C:\SJSURepo\Data260\Data260-2837\code\web_application\kafka_pipeline>python planner.py
2026-11-02 00:03:26.507 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_inspection_requests partitions=1
2026-11-02 00:03:26.508 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_inspection_responses partitions=1
2026-11-02 00:03:26.509 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_inbox partitions=1
2026-11-02 00:03:26.510 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_tasks partitions=2
2026-11-02 00:03:26.511 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_drafts partitions=1
2026-11-02 00:03:26.512 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_final partitions=1
2026-11-02 00:03:26.512 | Shashank Ranjan | s2837 | PLANNER-1   | topic ready s2837_dead_letter partitions=1
2026-11-02 00:03:26.527 | Shashank Ranjan | s2837 | PLANNER-1   | planner started pid=8120 topic=s2837_inbox group=s2837_
planners idempotency=True commit_mode=after_dlq=s2837_dead_letter max_retries=3
2026-11-02 00:03:27.535 | Shashank Ranjan | s2837 | PLANNER-1   | READY assigned partitions=[0] positions={0: 0} - Consum
er is ready and listening...
```

Writer is ready

```
PS C:\SJSURepo\Data260\Data260-2837\code\web_application\kafka_pipeline> python writer.py
2026-11-02 00:06:08.281 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_inspection_requests partitions=1
2026-11-02 00:06:08.281 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_inspection_responses partitions=1
2026-11-02 00:06:08.282 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_inbox partitions=1
2026-11-02 00:06:08.283 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_tasks partitions=2
2026-11-02 00:06:08.283 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_drafts partitions=1
2026-11-02 00:06:08.284 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_final partitions=1
2026-11-02 00:06:08.285 | Shashank Ranjan | s2837 | WRITER-1    | topic ready s2837_dead_letter partitions=1
2026-11-02 00:06:08.298 | Shashank Ranjan | s2837 | WRITER-1    | writer started pid=6712 topic=s2837_tasks group=s2837_w
riters idempotency=True commit_mode=after_dlq=s2837_dead_letter max_retries=3
2026-11-02 00:06:09.309 | Shashank Ranjan | s2837 | WRITER-1    | READY assigned partitions=[0, 1] positions={0: 0, 1: 0}
 - Consumer is ready and listening...
```

Reviewer is ready

```
PS C:\SJSURepo\Data260\Data260-2837\code\web_application\kafka_pipeline> python reviewer.py
2026-11-02 00:06:47.137 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_inspection_requests partitions=1
2026-11-02 00:06:47.137 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_inspection_responses partitions=1
2026-11-02 00:06:47.139 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_inbox partitions=1
2026-11-02 00:06:47.140 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_tasks partitions=2
2026-11-02 00:06:47.140 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_drafts partitions=1
2026-11-02 00:06:47.141 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_final partitions=1
2026-11-02 00:06:47.142 | Shashank Ranjan | s2837 | REVIEWER-1 | topic ready s2837_dead_letter partitions=1
2026-11-02 00:06:47.155 | Shashank Ranjan | s2837 | REVIEWER-1 | reviewer started pid=11864 topic=s2837_drafts group=s28
37_reviewers idempotency=True commit_mode=after dlq=s2837_dead_letter max_retries=3
2026-11-02 00:06:48.170 | Shashank Ranjan | s2837 | REVIEWER-1 | READY assigned partitions=[0] positions={0: 0} - Consum
er is ready and listening...
```

--send questions.py

```
PS C:\Users\Shashank Ranjan> cd \
PS C:\> cd .\SJSURepo\Data260\Data260-2837\code\web_application\kafka_pipeline\
PS C:\SJSURepo\Data260\Data260-2837\code\web_application\kafka_pipeline> python .\send_question.py
2026-11-02 00:11:55.806 | Shashank Ranjan | s2837 | SENDER     | ================================================
===============
2026-11-02 00:11:55.807 | Shashank Ranjan | s2837 | SENDER     | HW8 Part 2 - Shashank Ranjan - inbox -> planner -> writ
er -> reviewer -> final
2026-11-02 00:11:55.810 | Shashank Ranjan | s2837 | SENDER     | ================================================
===============
2026-11-02 00:11:55.958 | Shashank Ranjan | s2837 | SENDER     | SENT s2837_inbox[p0@0] correlation_id=c88dcfc7-8966-41c
3-8ba1-afb40890e891 question='A restaurant just received a WARNING status with a score of 72 in its health inspection. W
hat should the owner do before the re-inspection?'
2026-11-02 00:11:55.959 | Shashank Ranjan | s2837 | SENDER     | waiting for correlation_id=c88dcfc7-8966-41c3-8ba1-afb4
0890e891 on s2837_final ...
2026-11-02 00:12:27.495 | Shashank Ranjan | s2837 | SENDER     | ================================================
===============
2026-11-02 00:12:27.497 | Shashank Ranjan | s2837 | SENDER     | MESSAGE JOURNEY (from the trace carried inside the mess
ages)
2026-11-02 00:12:27.498 | Shashank Ranjan | s2837 | SENDER     | ================================================
===============
2026-11-02 00:12:27.498 | Shashank Ranjan | s2837 | SENDER     | you     -> s2837_inbox    at 2026-11-02T08:11:55.841+00
:00 correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
2026-11-02 00:12:27.501 | Shashank Ranjan | s2837 | SENDER     | planner  read s2837_inbox[p0@0] at 2026-11-02T08:11:55.
962+00:00 -> s2837_tasks at 2026-11-02T08:12:06.933+00:00 (planner-1, pid 8120)
2026-11-02 00:12:27.502 | Shashank Ranjan | s2837 | SENDER     | writer   read s2837_tasks[p0@0] at 2026-11-02T08:12:06.
947+00:00 -> s2837_drafts at 2026-11-02T08:12:27.459+00:00 (writer-1, pid 6712)
2026-11-02 00:12:27.503 | Shashank Ranjan | s2837 | SENDER     | reviewer read s2837_drafts[p0@0] at 2026-11-02T08:12:27
.478+00:00 -> s2837_final at 2026-11-02T08:12:27.481+00:00 (reviewer-1, pid 11864)
2026-11-02 00:12:27.504 | Shashank Ranjan | s2837 | SENDER     | correlation_id preserved through every stage: True
```

2.5 One complete message journey (inbox -> final)

```
00:11:55.958 SENDER      SENT s2837_inbox[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
question='A restaurant just received a WARNING status with a score ...
00:11:55.959 PLANNER-1   RECEIVED s2837_inbox[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
message={"correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e89...
00:11:55.962 PLANNER-1   PROCESSING correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
00:12:06.941 PLANNER-1   SENT s2837_tasks[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
message={"correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e891", ...
00:12:06.942 WRITER-1    RECEIVED s2837_tasks[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
message={"correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e89...
00:12:27.471 WRITER-1    SENT s2837_drafts[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
message={"correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e891",...
00:12:27.473 REVIEWER-1  RECEIVED s2837_drafts[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
message={"correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e8...
00:12:27.478 REVIEWER-1  PROCESSING correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
00:12:27.491 REVIEWER-1  SENT s2837_final[p0@0] correlation_id=c88dcfc7-8966-41c3-8ba1-afb40890e891
message={"correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e891", ...
```

Stage	Agent	Read from	Received	Finished
planner	planner-1	s2837_inbox[p0@0]	08:11:55.962	08:12:06.933
writer	writer-1	s2837_tasks[p0@0]	08:12:06.947	08:12:27.459
reviewer	reviewer-1	s2837_drafts[p0@0]	08:12:27.478	08:12:27.481

2.6 Final JSON from s2837_final


```
{
  "correlation_id": "c88dcfc7-8966-41c3-8ba1-afb40890e891",
  "status": "approved",
  "answer": "Before the re-inspection, the restaurant owner should take the following steps:\n\n1. **Review
the Inspection Report and Issues**: Carefully examine the inspection report to understand the specific
issues highlighted by the warning. This might include areas such as food safety practices, cleanliness,
food handling, and employee health.\n\n2. **Implement Corrective Actions**: Address the identified issues
immediately. This could involve:\n   - Improving food storage and preparation practices to ensure proper
temperature control.\n   - Enhancing cleaning procedures to eliminate cross-contamination.\n   - Ensuring
proper handwashing and glove use by staff.\n   - Training staff on food safety protocols and proper
handling of food.\n   - Implementing a system to monitor and log food temperatures and staff health.\n\n3.
**Schedule a Follow-Up Inspection**: Arrange for a follow-up inspection within a reasonable timeframe,
typically 1-2 weeks, to ensure the corrective measures are being effectively implemented. This will allow
the restaurant to demonstrate that the necessary improvements have been made and that the issues
highlighted in the warning have been resolved.\n\nBy following these steps, the restaurant can demonstrate
to the health department that it is taking the necessary steps to improve its food safety practices and
address the issues highlighted in the warning, thereby potentially reducing the warning status and
improving its overall health inspection score.",
  "review": {
    "checks": {
      "non_empty": true,
      "length_40_400_words": true,
      "on_domain": true,
      "no_error_text": true
    },
    "word_count": 209,
    "reviewer": "reviewer-1",
    "reviewed_at": "2026-11-02T08:12:27.479+00:00"
  },
  "writer": {
    "instance": "writer-1",
    "model": "ollama:qwen2.5:1.5b (temperature 0, seed 2837)",
    "written_at": "2026-11-02T08:12:27.459+00:00"
  }
}
```


2.7 How Kafka helped the agents coordinate

Kafka acted as the only contract between the agents: each agent knows just its input topic, its output topic and the JSON shape, so I could start, stop or restart any agent independently and the messages waited in the topic instead of being lost. Consumer groups and committed offsets let every agent resume exactly where it stopped, the two-partition tasks topic lets me add a second Writer without code changes, and the correlation_id plus the trace inside each message gave me an end-to-end audit of one question across three processes. The cost is that the pipeline is asynchronous: send_question.py has to wait on s2837_final and match the correlation_id instead of getting a return value.

Part 3 - Failure Behavior

All experiments are produced by run_hw08_failures.py, which starts the agents as separate processes with their own consumer groups and test_run tag, toggles the safety feature off (before) and on (after), and writes

raw/part3_*.json|csv|txt. Agents use the deterministic writer template here so timing reflects Kafka, not the model.

3.I Duplicate delivery - idempotency

```
# agent_base.py
def _process_once(self, record, where, attempts):
        message = decode(record.value)  # ValueError -> poison
        correlation_id = message.get("correlation_id")
        if not isinstance(correlation_id, str) or not correlation_id:
            raise PermanentError("missing correlation_id")
        if message.get("test_run") != self.args.test_run:
            log(self.component, f"IGNORED {where} test_run={message.get('test_run')} "
                                f"(this agent handles test_run={self.args.test_run})")
            return "ignored"
        if attempts == 1:
            log(self.component, f"RECEIVED {where} correlation_id={correlation_id} "
                                f"message={short(message, 300)}")
        if self.store is not None:
            status = self.store.status(correlation_id)
            if status == "done":
                log(self.component, f"DUPLICATE correlation_id={correlation_id} already "
                                    "processed - skipped, no new output (idempotent)")
                trace(self.component, "DUPLICATE_SKIPPED", correlation_id, where=where)
                return "duplicate_skipped"
            if status == "in_progress" and attempts == 1:
                log(self.component, f"REDELIVERY correlation_id={correlation_id} - previous "
                                    "attempt never finished (consumer died); processing again")
            self.store.mark_started(correlation_id)

        message.setdefault("trace", [])
        stage = {"stage": self.name, "agent": self.args.instance, "pid": os.getpid(),
                 "topic": record.topic, "partition": record.partition, "offset": record.offset,
                 "received_at": now_iso()}
        log(self.component, f"PROCESSING correlation_id={correlation_id}")
        outputs = self.handler(message, record, self)
        stage["finished_at"] = now_iso()
        for topic, value, key in outputs:
    # ... (see repository for the rest)
```

Run	Deliveries	Drafts produced	Finals produced	DUPLICATE skips
before	2	2	2	0
after	2	1	1	1

Before fix (--no-idempotency):

```
00:19:41.815 REVIEWER-DUP-BEFORE reviewer started pid=1672 topic=s2837_drafts
group=s2837_dup_r_before_20261102001941 idempotency=False commit_mode=after dlq=s2837_dea...
00:19:41.815 WRITER-DUP-BEFORE writer started pid=13752 topic=s2837_tasks
group=s2837_dup_w_before_20261102001941 idempotency=False commit_mode=after dlq=s2837_dead_le...
00:19:44.038 WRITER-DUP-BEFORE RECEIVED s2837_tasks[p0@1] correlation_id=7bd62cad-a1cc-4eb3-beae-
ada50482b07c message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ada50...
00:19:44.051 WRITER-DUP-BEFORE SENT s2837_drafts[p0@1] correlation_id=7bd62cad-a1cc-4eb3-beae-ada50482b07c
message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ada50482...
00:19:44.052 REVIEWER-DUP-BEFORE RECEIVED s2837_drafts[p0@1] correlation_id=7bd62cad-a1cc-4eb3-beae-
ada50482b07c message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ad...
```

```
00:19:44.057 WRITER-DUP-BEFORE COMMITTED offset 2 on p0 after success
00:19:44.067 REVIEWER-DUP-BEFORE SENT s2837_final[p0@1] correlation_id=7bd62cad-a1cc-4eb3-beae-ada50482b07c
message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ada5048...
00:19:44.071 REVIEWER-DUP-BEFORE COMMITTED offset 2 on p0 after success
00:19:45.149 WRITER-DUP-BEFORE RECEIVED s2837_tasks[p0@2] correlation_id=7bd62cad-a1cc-4eb3-beae-
ada50482b07c message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ada50...
00:19:45.184 WRITER-DUP-BEFORE SENT s2837_drafts[p0@2] correlation_id=7bd62cad-a1cc-4eb3-beae-ada50482b07c
message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ada50482...
00:19:45.185 REVIEWER-DUP-BEFORE RECEIVED s2837_drafts[p0@2] correlation_id=7bd62cad-a1cc-4eb3-beae-
ada50482b07c message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ad...
00:19:45.191 WRITER-DUP-BEFORE COMMITTED offset 3 on p0 after success
00:19:45.203 REVIEWER-DUP-BEFORE SENT s2837_final[p0@2] correlation_id=7bd62cad-a1cc-4eb3-beae-ada50482b07c
message={"correlation_id": "7bd62cad-a1cc-4eb3-beae-ada5048...
00:19:45.208 REVIEWER-DUP-BEFORE COMMITTED offset 3 on p0 after success
```

After fix (SQLite idempotency store keyed by correlation_id):

```
00:19:55.625 WRITER-DUP-AFTER writer started pid=6620 topic=s2837_tasks
group=s2837_dup_w_after_20261102001941 idempotency=True commit_mode=after dlq=s2837_dead_letter...
00:19:55.628 REVIEWER-DUP-AFTER reviewer started pid=20912 topic=s2837_drafts
group=s2837_dup_r_after_20261102001941 idempotency=True commit_mode=after dlq=s2837_dead_...
00:19:57.736 WRITER-DUP-AFTER RECEIVED s2837_tasks[p1@0] correlation_id=952a972d-2171-4704-a484-
ba8e6f5d210b message={"correlation_id": "952a972d-2171-4704-a484-ba8e6f...
00:19:57.747 REVIEWER-DUP-AFTER RECEIVED s2837_drafts[p0@3] correlation_id=952a972d-2171-4704-a484-
ba8e6f5d210b message={"correlation_id": "952a972d-2171-4704-a484-ba8...
00:19:57.747 WRITER-DUP-AFTER SENT s2837_drafts[p0@3] correlation_id=952a972d-2171-4704-a484-ba8e6f5d210b
message={"correlation_id": "952a972d-2171-4704-a484-ba8e6f5d2...
00:19:57.754 WRITER-DUP-AFTER COMMITTED offset 1 on p1 after success
00:19:57.760 REVIEWER-DUP-AFTER SENT s2837_final[p0@3] correlation_id=952a972d-2171-4704-a484-ba8e6f5d210b
message={"correlation_id": "952a972d-2171-4704-a484-ba8e6f5d...
00:19:57.765 REVIEWER-DUP-AFTER COMMITTED offset 4 on p0 after success
00:19:58.852 WRITER-DUP-AFTER RECEIVED s2837_tasks[p1@1] correlation_id=952a972d-2171-4704-a484-
ba8e6f5d210b message={"correlation_id": "952a972d-2171-4704-a484-ba8e6f...
00:19:58.856 WRITER-DUP-AFTER DUPLICATE correlation_id=952a972d-2171-4704-a484-ba8e6f5d210b already
processed - skipped, no new output (idempotent)
00:19:58.864 WRITER-DUP-AFTER COMMITTED offset 2 on p1 after duplicate_skipped
```

3.II Consumer death - commit offsets only after successful processing

```
# agent_base.py
def _handle(self, record):
        where = f"{record.topic}[p{record.partition}@{record.offset}]"
        if self.args.commit_mode == "early":
            self.consumer.commit()
            log(self.component, f"COMMITTED offset {record.offset + 1} on p{record.partition} "
                                "BEFORE processing (unsafe early-commit mode)")
        attempts = 0
        while True:
            attempts += 1
            try:
                outcome = self._process_once(record, where, attempts)
                break
            except Exception as exc:  # retry / dead-letter
                permanent = isinstance(exc, (PermanentError, ValueError, UnicodeDecodeError))
                log(self.component, f"ERROR attempt {attempts}"
                                    f"{'' if self.args.no_dlq else '/' + str(self.args.max_retries)}"
                                    f" at {where}: {type(exc).__name__}: {exc}")
```

```
            if not self.args.no_dlq and attempts >= self.args.max_retries:
                self._dead_letter(record, exc, attempts, permanent)
                outcome = "dead_letter"
                break
            backoff = 1.0 if self.args.no_dlq else min(0.5 * 2 ** (attempts - 1), 4)
            log(self.component, f"RETRY in {backoff:.1f}s (message blocks this partition "
                                "until it succeeds" + (")" if self.args.no_dlq
                                                       else " or reaches the limit)"))
            time.sleep(backoff)
        if self.args.commit_mode == "after":
            self.consumer.commit()
            log(self.component, f"COMMITTED offset {record.offset + 1} on p{record.partition} "
                                f"after {outcome}")
```

Run	commit mode	msg offset	committed at kill	after restart	redelivered	lost
before	early	p1@3	{'1': 4, '0': 3}	{'1': 4, '0': 3}	False	True
after	after	p1@5	{'1': 5, '0': 3}	{'1': 6, '0': 3}	True	False

Before fix (--commit-mode early, offset committed on receipt):

```
00:20:09.289 WRITER-CRASH-BEFORE writer started pid=8676 topic=s2837_tasks
group=s2837_crash_before_20261102001941 idempotency=True commit_mode=early dlq=s2837_dead_le...
00:20:10.298 WRITER-CRASH-BEFORE READY assigned partitions=[0, 1] positions={0: 3, 1: 2} - Consumer is
ready and listening...
00:20:11.393 WRITER-CRASH-BEFORE COMMITTED offset 3 on p1 BEFORE processing (unsafe early-commit mode)
00:20:11.394 WRITER-CRASH-BEFORE RECEIVED s2837_tasks[p1@2] correlation_id=9655658f-b946-4f06-a55b-
13526eecc60b message={"correlation_id": "9655658f-b946-4f06-a55b-135...
00:20:11.408 WRITER-CRASH-BEFORE SENT s2837_drafts[p0@4] correlation_id=9655658f-b946-4f06-a55b-
13526eecc60b message={"correlation_id": "9655658f-b946-4f06-a55b-13526e...
00:20:11.504 WRITER-CRASH-BEFORE COMMITTED offset 4 on p1 BEFORE processing (unsafe early-commit mode)
00:20:11.506 WRITER-CRASH-BEFORE RECEIVED s2837_tasks[p1@3] correlation_id=8f9990cc-95db-49df-a697-
707ba1616398 message={"correlation_id": "8f9990cc-95db-49df-a697-707...
00:20:11.511 WRITER-CRASH-BEFORE WORKING slowly for 8s on correlation_id=8f9990cc-95db-49df-a697-
707ba1616398 (offset not committed yet - kill me now)
00:20:16.078 WRITER-CRASH-BEFORE-RESTART writer started pid=21408 topic=s2837_tasks
group=s2837_crash_before_20261102001941 idempotency=True commit_mode=early dlq=s283...
00:20:17.082 WRITER-CRASH-BEFORE-RESTART READY assigned partitions=[] positions={} - Consumer is ready and
listening...
00:20:23.117 WRITER-CRASH-BEFORE-RESTART READY assigned partitions=[0, 1] positions={0: 3, 1: 4} - Consumer
is ready and listening...
```

After fix (commit after the draft is acked):

```
00:21:05.181 WRITER-CRASH-AFTER writer started pid=8244 topic=s2837_tasks
group=s2837_crash_after_20261102001941 idempotency=True commit_mode=after dlq=s2837_dead_lett...
00:21:06.191 WRITER-CRASH-AFTER READY assigned partitions=[0, 1] positions={0: 3, 1: 4} - Consumer is ready
and listening...
00:21:07.267 WRITER-CRASH-AFTER RECEIVED s2837_tasks[p1@4] correlation_id=ef5d7b46-f974-4042-bccc-
586ee9904950 message={"correlation_id": "ef5d7b46-f974-4042-bccc-586e...
00:21:07.281 WRITER-CRASH-AFTER SENT s2837_drafts[p0@5] correlation_id=ef5d7b46-f974-4042-bccc-586ee9904950
message={"correlation_id": "ef5d7b46-f974-4042-bccc-586ee99...
00:21:07.287 WRITER-CRASH-AFTER COMMITTED offset 5 on p1 after success
00:21:07.377 WRITER-CRASH-AFTER RECEIVED s2837_tasks[p1@5] correlation_id=c2d41f18-3aba-4060-b2d0-
8de80dd3349d message={"correlation_id": "c2d41f18-3aba-4060-b2d0-8de8...
```

```
00:21:07.385 WRITER-CRASH-AFTER WORKING slowly for 8s on correlation_id=c2d41f18-3aba-4060-b2d0-
8de80dd3349d (offset not committed yet - kill me now)
00:21:11.913 WRITER-CRASH-AFTER-RESTART writer started pid=22656 topic=s2837_tasks
group=s2837_crash_after_20261102001941 idempotency=True commit_mode=after dlq=s2837_...
00:21:12.914 WRITER-CRASH-AFTER-RESTART READY assigned partitions=[] positions={} - Consumer is ready and
listening...
00:21:18.437 WRITER-CRASH-AFTER-RESTART READY assigned partitions=[0, 1] positions={0: 3, 1: 5} - Consumer
is ready and listening...
00:21:18.439 WRITER-CRASH-AFTER-RESTART RECEIVED s2837_tasks[p1@5] correlation_id=c2d41f18-3aba-4060-b2d0-
8de80dd3349d message={"correlation_id": "c2d41f18-3aba-4060-b...
00:21:18.440 WRITER-CRASH-AFTER-RESTART REDELIVERY correlation_id=c2d41f18-3aba-4060-b2d0-8de80dd3349d -
previous attempt never finished (consumer died); processing again
00:21:18.445 WRITER-CRASH-AFTER-RESTART WORKING slowly for 8s on correlation_id=c2d41f18-3aba-4060-b2d0-
8de80dd3349d (offset not committed yet - kill me now)
00:21:26.461 WRITER-CRASH-AFTER-RESTART SENT s2837_drafts[p0@6] correlation_id=c2d41f18-3aba-4060-b2d0-
8de80dd3349d message={"correlation_id": "c2d41f18-3aba-4060-b2d0...
00:21:26.467 WRITER-CRASH-AFTER-RESTART COMMITTED offset 6 on p1 after success
```

3.III Poison message - bounded retries and s2837_dead_letter

```
# agent_base.py
def _dead_letter(self, record, exc, attempts, permanent):
        raw = record.value.decode("utf-8", "replace") if record.value else None
        try:
            correlation_id = decode(record.value).get("correlation_id")
        except Exception:
            correlation_id = None
        letter = {
            "correlation_id": correlation_id,
            "failed_stage": self.name,
            "failed_agent": self.args.instance,
            "source_topic": record.topic, "source_partition": record.partition,
            "source_offset": record.offset,
            "error_type": type(exc).__name__, "error": str(exc),
            "permanent": permanent, "attempts": attempts,
            "raw_value": raw, "dead_lettered_at": now_iso(),
            "traceback_tail": traceback.format_exception_only(type(exc), exc)[-1].strip(),
        }
        meta = send_json(self.producer, cfg.DEAD_LETTER_TOPIC, letter, key=correlation_id)
        log(self.component, f"DEAD-LETTER after {attempts} attempts -> "
                            f"{cfg.DEAD_LETTER_TOPIC}[p{meta.partition}@{meta.offset}] "
                            f"source={record.topic}[p{record.partition}@{record.offset}]")
        trace(self.component, "DEAD_LETTER", correlation_id, offset=record.offset)
```

Run	Attempts	Good msgs processed	Dead-letter records
before	18	0/3	0
after	3	3/3	1

Record in s2837_dead_letter:

```
{
  "correlation_id": "dc3877fa-137d-4c63-ad8b-0a96fee03cbc",
  "failed_stage": "writer",
  "failed_agent": "writer-poison-after",
```

```
  "source_topic": "s2837_tasks",
  "source_partition": 1,
  "source_offset": 10,
  "error_type": "PermanentError",
  "error": "plan.steps must be a non-empty list of strings",
  "permanent": true,
  "attempts": 3,
  "raw_value": "{\"correlation_id\": \"dc3877fa-137d-4c63-ad8b-0a96fee03cbc\", \"test_run\": \"poison-
after-20261102001941\", \"kind\": \"answer_task\", \"question\": \"poison\", \"plan\": \"THIS-IS-NOT-A-
LIST\"}",
  "dead_lettered_at": "2026-11-02T08:21:50.120+00:00",
  "traceback_tail": "web_application.kafka_pipeline.agent_base.PermanentError: plan.steps must be a non-
empty list of strings"
}
```


Before fix (--no-dlq, retries forever):

```
00:21:28.965 WRITER-POISON-BEFORE RECEIVED s2837_tasks[p1@6] correlation_id=9484baec-c2d7-49f2-88c2-
d4e572c3abf4 message={"correlation_id": "9484baec-c2d7-49f2-88c2-d4...
00:21:28.969 WRITER-POISON-BEFORE ERROR attempt 1 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
00:21:28.969 WRITER-POISON-BEFORE RETRY in 1.0s (message blocks this partition until it succeeds)
00:21:29.976 WRITER-POISON-BEFORE ERROR attempt 2 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
00:21:29.979 WRITER-POISON-BEFORE RETRY in 1.0s (message blocks this partition until it succeeds)
00:21:30.988 WRITER-POISON-BEFORE ERROR attempt 3 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
00:21:30.990 WRITER-POISON-BEFORE RETRY in 1.0s (message blocks this partition until it succeeds)
00:21:31.996 WRITER-POISON-BEFORE ERROR attempt 4 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
00:21:31.999 WRITER-POISON-BEFORE RETRY in 1.0s (message blocks this partition until it succeeds)
00:21:33.006 WRITER-POISON-BEFORE ERROR attempt 5 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
00:21:33.008 WRITER-POISON-BEFORE RETRY in 1.0s (message blocks this partition until it succeeds)
00:21:34.018 WRITER-POISON-BEFORE ERROR attempt 6 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
00:21:34.020 WRITER-POISON-BEFORE RETRY in 1.0s (message blocks this partition until it succeeds)
00:21:35.032 WRITER-POISON-BEFORE ERROR attempt 7 at s2837_tasks[p1@6]: PermanentError: plan.steps must be
a non-empty list of strings
```

After fix:

```
00:21:48.599 WRITER-POISON-AFTER RECEIVED s2837_tasks[p1@10] correlation_id=dc3877fa-137d-4c63-ad8b-
0a96fee03cbc message={"correlation_id": "dc3877fa-137d-4c63-ad8b-0a...
00:21:48.604 WRITER-POISON-AFTER ERROR attempt 1/3 at s2837_tasks[p1@10]: PermanentError: plan.steps must
be a non-empty list of strings
00:21:48.605 WRITER-POISON-AFTER RETRY in 0.5s (message blocks this partition until it succeeds or reaches
the limit)
00:21:49.109 WRITER-POISON-AFTER ERROR attempt 2/3 at s2837_tasks[p1@10]: PermanentError: plan.steps must
be a non-empty list of strings
00:21:49.110 WRITER-POISON-AFTER RETRY in 1.0s (message blocks this partition until it succeeds or reaches
the limit)
00:21:50.117 WRITER-POISON-AFTER ERROR attempt 3/3 at s2837_tasks[p1@10]: PermanentError: plan.steps must
be a non-empty list of strings
00:21:50.140 WRITER-POISON-AFTER DEAD-LETTER after 3 attempts -> s2837_dead_letter[p0@0]
source=s2837_tasks[p1@10]
00:21:50.148 WRITER-POISON-AFTER COMMITTED offset 11 on p1 after dead_letter
```

```
00:21:50.151 WRITER-POISON-AFTER RECEIVED s2837_tasks[p1@11] correlation_id=3943cfc8-03d1-4150-8965-
fa8cd17cb558 message={"correlation_id": "3943cfc8-03d1-4150-8965-fa...
00:21:50.264 WRITER-POISON-AFTER SENT s2837_drafts[p0@7] correlation_id=3943cfc8-03d1-4150-8965-
fa8cd17cb558 message={"correlation_id": "3943cfc8-03d1-4150-8965-fa8cd1...
00:21:50.278 WRITER-POISON-AFTER COMMITTED offset 12 on p1 after success
00:21:50.281 WRITER-POISON-AFTER RECEIVED s2837_tasks[p1@12] correlation_id=7ce400f7-a796-46c5-bff1-
2c76344dc23e message={"correlation_id": "7ce400f7-a796-46c5-bff1-2c...
00:21:50.292 WRITER-POISON-AFTER SENT s2837_drafts[p0@8] correlation_id=7ce400f7-a796-46c5-bff1-
2c76344dc23e message={"correlation_id": "7ce400f7-a796-46c5-bff1-2c7634...
00:21:50.300 WRITER-POISON-AFTER COMMITTED offset 13 on p1 after success
00:21:50.301 WRITER-POISON-AFTER RECEIVED s2837_tasks[p1@13] correlation_id=7f24831e-3276-42d7-881c-
5e6de06cb43d message={"correlation_id": "7f24831e-3276-42d7-881c-5e...
00:21:50.308 WRITER-POISON-AFTER SENT s2837_drafts[p0@9] correlation_id=7f24831e-3276-42d7-881c-
5e6de06cb43d message={"correlation_id": "7f24831e-3276-42d7-881c-5e6de0...
00:21:50.315 WRITER-POISON-AFTER COMMITTED offset 14 on p1 after success
```

3.IV Ordering - 2 partitions, 2 Writers in one group, VERIFY_SEED messages

```
# run_hw08_failures.py
def ordering_messages():
    rng = random.Random(cfg.VERIFY_SEED)
    entities = [f"INSP-{cfg.SID4}-{c}" for c in "ABCD"]
    lifecycle = ["SCHEDULED", "STARTED", "VIOLATION_RECORDED", "SCORE_UPDATED",
                 "STATUS_CHANGED", "REINSPECTION_BOOKED", "CLOSED", "NOTE_ADDED"]
    seq = {e: 0 for e in entities}
    out = []
    for i in range(20):
        entity = rng.choice(entities)
        seq[entity] += 1
        out.append({"global_index": i, "entity_id": entity, "seq": seq[entity],
                    "event": lifecycle[(seq[entity] - 1) % len(lifecycle)],
                    "work_ms": rng.randint(30, 900)})
    return out

# partition key:
meta = send_json(producer, cfg.TASKS_TOPIC, body, key=msg['entity_id'])  # keyed run
```

Run - unkeyed (no key):

Entity	Processed seq order	Partitions	In order
INSP-2837-A	[2, 1, 3]	[0, 1]	False
INSP-2837-B	[1, 3, 2, 4, 5, 6, 7, 8]	[0, 1]	False
INSP-2837-C	[1, 2, 3, 4, 5]	[0, 1]	True
INSP-2837-D	[1, 2, 3, 4]	[0, 1]	True

```
00:21:53.794 WRITER-ORDER-1-UNKEYED READY assigned partitions=[0] positions={0: 3} - Consumer is ready and
listening...
00:21:53.794 WRITER-ORDER-2-UNKEYED READY assigned partitions=[1] positions={1: 14} - Consumer is ready and
listening...
00:21:54.949 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-D seq=1 partition=0 offset=3
00:21:54.983 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-C seq=1 partition=1 offset=14
00:21:55.708 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-B seq=1 partition=0 offset=4
00:21:55.736 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-D seq=2 partition=1 offset=15
00:21:55.820 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-D seq=3 partition=0 offset=5
00:21:55.905 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-B seq=3 partition=0 offset=6
```

```
00:21:55.954 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-B seq=2 partition=1 offset=16
00:21:56.685 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-B seq=4 partition=1 offset=17
00:21:56.697 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-B seq=5 partition=0 offset=7
00:21:56.731 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-C seq=2 partition=1 offset=18
00:21:57.365 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-B seq=6 partition=1 offset=19
00:21:57.449 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-A seq=2 partition=1 offset=20
00:21:57.581 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-C seq=3 partition=0 offset=8
00:21:58.050 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-B seq=7 partition=1 offset=21
00:21:58.167 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-A seq=1 partition=0 offset=9
00:21:58.371 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-D seq=4 partition=1 offset=22
00:21:58.459 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-C seq=4 partition=0 offset=10
00:21:58.688 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-A seq=3 partition=0 offset=11
00:21:58.991 WRITER-ORDER-2-UNKEYED ORDER entity=INSP-2837-B seq=8 partition=1 offset=23
00:21:59.075 WRITER-ORDER-1-UNKEYED ORDER entity=INSP-2837-C seq=5 partition=0 offset=12
```

Run - keyed (key = entity_id):

Entity	Processed seq order	Partitions	In order
INSP-2837-A	[1, 2, 3]	[1]	True
INSP-2837-B	[1, 2, 3, 4, 5, 6, 7, 8]	[0]	True
INSP-2837-C	[1, 2, 3, 4, 5]	[0]	True
INSP-2837-D	[1, 2, 3, 4]	[1]	True

```
00:22:00.596 WRITER-ORDER-1-KEYED READY assigned partitions=[0] positions={0: 13} - Consumer is ready and
listening...
00:22:00.596 WRITER-ORDER-2-KEYED READY assigned partitions=[1] positions={1: 24} - Consumer is ready and
listening...
00:22:01.852 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-D seq=1 partition=1 offset=24
00:22:01.886 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-C seq=1 partition=0 offset=13
00:22:02.608 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-D seq=2 partition=1 offset=25
00:22:02.643 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=1 partition=0 offset=14
00:22:02.720 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-D seq=3 partition=1 offset=26
00:22:02.860 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=2 partition=0 offset=15
00:22:02.966 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=3 partition=0 offset=16
00:22:03.293 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-A seq=1 partition=1 offset=27
00:22:03.375 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-A seq=2 partition=1 offset=28
00:22:03.609 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-A seq=3 partition=1 offset=29
00:22:03.715 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=4 partition=0 offset=17
00:22:03.951 WRITER-ORDER-2-KEYED ORDER entity=INSP-2837-D seq=4 partition=1 offset=30
00:22:04.507 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=5 partition=0 offset=18
00:22:04.561 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-C seq=2 partition=0 offset=19
00:22:05.453 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-C seq=3 partition=0 offset=20
00:22:06.107 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=6 partition=0 offset=21
00:22:06.379 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-C seq=4 partition=0 offset=22
00:22:06.992 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=7 partition=0 offset=23
00:22:07.360 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-C seq=5 partition=0 offset=24
00:22:07.978 WRITER-ORDER-1-KEYED ORDER entity=INSP-2837-B seq=8 partition=0 offset=25
```

3.V Failure table

Failure scenario	Before fix	After fix
Duplicate delivery	same correlation_id delivered twice -> 2 drafts, 2 finals (extra side effect)	idempotency store -> 1 draft, 1 final, 1 DUPLICATE skips
Consumer killed mid-work	commit-before-processing: offset already {1: 4, 0: 3} at kill -> after restart	commit-after-success: committed {1: 5, 0: 3} at kill -> redelivered, REDELIVERY

	message_lost=True	detected=True, draft completed=True, committed now {1: 6, 0: 3}
Poison message	unbounded retry: 18 attempts in 15s, good messages processed 0/3 (partition blocked)	3 attempts (limit 3) -> 1 record in s2837_dead_letter, good messages processed 3/3
Two partitions/two consumers	no key: 4/4 entities split across partitions, 2/4 processed out of order (2 inversions)	key=entity_id: 0/4 split, 0/4 out of order (0 inversions)

3.VI Comparison with the HW2 in-process graph

In HW2 my LangGraph Planner -> Reviewer -> Supervisor graph ran as function calls in one Python process: each node returned its state directly to the next node.

Benefits gained by moving to Kafka:

- Independent scaling - a second Writer joined the same consumer group and took one of the two s2837_tasks partitions with no code change (Part 3.IV).
- Decoupled failure - when the Writer was killed, the Planner and Reviewer kept running and the unfinished task stayed in Kafka until the restarted Writer finished it (Part 3.II); in HW2 an exception in one node stopped the whole graph run.

New responsibilities I now handle myself:

- Duplicate delivery - Kafka is at-least-once, so the same correlation_id can arrive twice; I added an idempotency store and offset commits after success (Part 3.I/II).
- Ordering and poison messages - order is only guaranteed inside one partition, so I had to choose entity_id as the message key; and a bad message would block a partition forever, so I added bounded retries and s2837_dead_letter (Part 3.III/IV).

verification.json (make verify-hw08)

---Final Verifiction (All 18 of 18 passed)

```
d
PASS  replayed_task_produces_no_second_draft
PASS  same_key_messages_land_in_one_partition
PASS  idempotency_store_tracks_in_progress_and_done

18/18 checks passed -> C:\SJSURepo\Data260\Data260-2837\reports\hw08\verification.json

C:\SJSURepo\Data260\Data260-2837\code>
```


```
"checks": [
  {
    "check": "kafka_reachable",
    "status": "PASS"
  },
  {
    "check": "all_prefix_topics_exist_with_partitions",
    "status": "PASS"
```

```
  },
  {
    "check": "create_then_get_roundtrip_through_kafka",
    "status": "PASS"
  },
  {
    "check": "correlation_id_preserved_request_to_response",
    "status": "PASS"
  },
  {
    "check": "request_has_required_envelope_fields",
    "status": "PASS"
  },
  {
    "check": "get_unknown_id_returns_error_with_correlation_id",
    "status": "PASS"
  },
  {
    "check": "unknown_operation_returns_error",
    "status": "PASS"
  },
  {
    "check": "invalid_or_missing_fields_rejected",
    "status": "PASS"
  },
  {
    "check": "duplicate_request_same_correlation_id_creates_no_second_entity",
    "status": "PASS"
  },
  {
    "check": "fastapi_running_on_port_base",
    "status": "PASS"
  },
  {
    "check": "fastapi_create_get_via_kafka",
    "status": "PASS"
  },
  {
    "check": "pipeline_inbox_to_final_with_status_and_answer",
    "status": "PASS"
  },
  {
    "check": "pipeline_correlation_id_preserved_all_stages",
    "status": "PASS"
  },
  {
    "check": "poison_message_retried_then_dead_lettered",
    "status": "PASS"
  },
  {
```

```
    "check": "message_after_poison_still_processed",
    "status": "PASS"
   },
   {
    "check": "replayed_task_produces_no_second_draft",
    "status": "PASS"
   },
   {
    "check": "same_key_messages_land_in_one_partition",
    "status": "PASS"
   },
   {
    "check": "idempotency_store_tracks_in_progress_and_done",
    "status": "PASS"
   }
  ],
  "summary": {
   "passed": 18,
   "total": 18
  }
}
```


AI Use

1. What I used an AI assistant for and what I did myself: I used Claude to draft the Kafka code for this homework (worker, request client, FastAPI wrapper, Planner/Writer/Reviewer agents, the failure-test runner, docker-compose, the smoke test and the report builder) on top of my existing HW1-HW7 repository. I did myself: computing my configuration (SID4 2837, PREFIX s2837, PORT_BASE 8137, VERIFY_SEED 262837), choosing Inspection as the primary entity and reusing my InspectionCreate schema, running Docker/Kafka and every experiment on my machine, taking the screenshots, checking each log and table against the raw files, and the final review of the report.

2. One AI-produced output that was wrong: the first version of the duplicate-delivery experiment reported 3 drafts for one replayed correlation_id instead of 2, which would have overstated the duplicate problem in the failure table.

3. How I detected it: I read every record on s2837_drafts with its writer instance and test tag. The extra draft came from my normal Part 2 writer (group s2837_writers), which was still running and also consumed the test message, because each consumer group receives every message on the topic.

4. What I changed and why it works now: every Part 3 test message now carries a test_run tag. Agents only handle messages whose tag matches their --test-run value and log IGNORED for everything else, and the runner counts only tagged outputs. Each experiment also uses its own consumer group. A rerun gave exactly 2 drafts

before the fix and 1 after it. The same check (replayed_task_produces_no_second_draft) is part of verify_hw08.py.